

# Tyler A. Chang

Research Scientist, Google DeepMind

he/him/his

tachang@ucsd.edu / tylerchang@google.com

tylerachang.github.io

---

## EDUCATION

---

- |      |                                                                                                                                                                                                                                                                         |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | <b>University of California San Diego</b> , La Jolla, CA<br>MS, PhD in Cognitive Science<br>Halicioğlu Data Science Institute, Graduate Fellow<br>Dissertation: <i>N-gram learning and pretraining dynamics in Transformer language models.</i>                         |
| 2020 | <b>Carleton College</b> , Northfield, MN<br>BA in Mathematics, summa cum laude<br>Thesis: <i>Topology of second order tensor fields.</i><br>BA in Cognitive Science, with distinction<br>Thesis: <i>Emergence of hierarchical syntax in neural machine translation.</i> |
| 2018 | <b>Eötvös Loránd University</b> , Budapest, Hungary<br>Budapest Semesters in Mathematics, fall 2018                                                                                                                                                                     |
| 2018 | <b>Doshisha University</b> , Kyoto, Japan<br>Carleton Linguistics and Culture Program, spring 2018                                                                                                                                                                      |

---

## RESEARCH FOCUS

---

**Large language models:** multilinguality, pretraining dynamics, mechanistic interpretability.  
**Cognitive linguistics:** distributional semantics, language acquisition.

---

## INDUSTRY EXPERIENCE

---

- |            |                                                                                                                                          |
|------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 2025-Pres. | <b>Google DeepMind</b> , Research Scientist<br>Gemini multilinguality and internationalization.                                          |
| 2022-2024  | <b>Google DeepMind and Google Research</b> , PhD Student Researcher<br>Pretraining data attribution and hallucination detection in LLMs. |
| 2021, 2022 | <b>Amazon Science</b> , Applied Scientist Intern, Alexa AI, AWS AI Labs                                                                  |
| 2019, 2020 | <b>Google</b> , Software Engineering Intern, Google Earth, Translate                                                                     |
| 2018       | <b>Google</b> , Engineering Practicum Intern, Chrome                                                                                     |

---

## PUBLICATIONS

*Peer-reviewed.*

*Note: top-tier venues in natural language processing are often conference proceedings rather than journals.*

- |      |                                                                                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | <b>Chang, T. A.</b> , Arnett, C., Tu, Z., & Bergen, B. K. Goldfish: Monolingual language models for 350 languages. <i>Language Resources and Evaluation Conference (LREC).</i> |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- 2025 **Chang, T. A.**, & Bergen, B. K. Bigram subnetworks: Mapping to next tokens in Transformer language models. *Conference on Neural Information Processing Systems* (NeurIPS). Spotlight paper.
- 2025 Arnett, C., **Chang, T. A.**, Biderman, S., & Bergen, B. K. Explaining and mitigating crosslingual tokenizer inequities. *Conference on Neural Information Processing Systems* (NeurIPS).
- 2025 **Chang, T. A.**, Rajagopal, D., Bolukbasi, T., Dixon, L., & Tenney, I. Scalable influence and fact tracing for large language model pretraining. *International Conference on Learning Representations* (ICLR).
- 2025 Arnett, C., **Chang, T. A.**, Michaelov, J., & Bergen, B. K. On the acquisition of shared grammatical representations in bilingual language models. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics* (ACL).
- 2024 **Chang, T. A.**, Tu, Z., & Bergen, B. K. Characterizing learning curves during language model pre-training: Learning, forgetting, and stability. *Transactions of the Association for Computational Linguistics* (TACL).
- 2024 **Chang, T. A.**, Arnett, C., Tu, Z., & Bergen, B. K. When is multilinguality a curse? Language modeling for 250 high- and low-resource languages. *Proceedings of the Conference on Empirical Methods in Natural Language Processing* (EMNLP). Received an outstanding paper award.
- 2024 **Chang, T. A.**, & Bergen, B. K. Language model behavior: A comprehensive survey. *Computational Linguistics*.
- 2024 Unger, L., **Chang, T. A.**, Savic, O., Bergen, B. K., & Sloutsky, V. M. When is a word in good company for learning? *Developmental Science*.
- 2024 **Chang, T. A.\***, Tomanek, K.\*, Hoffmann, J., Thain, N., van Liemt, E., Meier-Hellstern, K., & Dixon, L. Detecting hallucination and coverage errors in retrieval augmented generation for controversial topics. *Proceedings of the Joint International Conference on Computational Linguistics, Language Resources, and Evaluation* (LREC-COLING). \*Equal contribution.
- 2024 Shah, C.◇, Chandak, Y.◇, Mane, A.◇, Bergen, B. K., & **Chang, T. A.** Correlations between multilingual language model geometry and crosslingual transfer performance. *Proceedings of the Joint International Conference on Computational Linguistics, Language Resources, and Evaluation* (LREC-COLING). ◇Undergraduate mentees.
- 2023 Michaelov, J.\*, Arnett, C.\*, **Chang, T. A.**, & Bergen, B. K. Structural priming demonstrates abstract grammatical representations in multilingual language models. *Proceedings of the Conference on Empirical Methods in Natural Language Processing* (EMNLP). \*Equal contribution.
- 2023 **Chang, T. A.**, Halder, K., Anna John, N., Vyas, Y., Benajiba, Y., Ballesteros, M., & Roth, D. Characterizing and measuring linguistic dataset drift. *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics* (ACL).
- 2023 Trott, S.\*, Jones, C. R.\*, **Chang, T. A.**, Michaelov, J., & Bergen, B. K. Do large language models know what humans know? *Cognitive Science*. \*Equal contribution.
- 2022 **Chang, T. A.**, Tu, Z., & Bergen, B. K. The geometry of multilingual language model representations. *Proceedings of the Conference on Empirical Methods in Natural Language Processing* (EMNLP).

- 2022 **Chang, T. A.**, & Bergen, B. K. Word acquisition in neural language models. *Transactions of the Association for Computational Linguistics (TACL)*. Presented at ACL 2022.
- 2022 **Chang, T. A.**, & Bergen, B. K. Does contextual diversity hinder early word acquisition? *Proceedings of the 44th Annual Conference of the Cognitive Science Society (CogSci)*.
- 2022 Jones, C. R., **Chang, T. A.**, Coulson, S., Michaelov, J., Trott, S., & Bergen, B. K. Distributional semantics still can't account for affordances. *Proceedings of the 44th Annual Conference of the Cognitive Science Society (CogSci)*.
- 2021 **Chang, T. A.**, Xu, Y., Xu, W., & Tu, Z. Convolutions and self-attention: Re-interpreting relative positions in pre-trained language models. *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (ACL-IJCNLP)*.
- 2021 Xu, Y., Xu, W., **Chang, T. A.**, & Tu, Z. Co-scale conv-attentional image transformers. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.

---

## PREPRINTS

---

- 2025 **Chang, T. A.\***, Arnett, C.\*, ... [300+ authors] ..., & Adelani, D. Global PIQA: Evaluating physical commonsense reasoning across 100+ languages and cultures. \*Equal contribution.
- Co-led a large scale benchmark covering over 100 languages, in collaboration with 300+ researchers from over 65 countries around the world.

---

## WORKSHOP PAPERS AND ABSTRACTS

---

*Peer-reviewed.*

- 2024 Arnett, C.\*, **Chang, T. A.\***, & Bergen, B. K. A bit of a problem: Measurement disparities in dataset sizes across languages. *Proceedings of the Annual Meeting of the Special Interest Group on Under-Resourced Languages (workshop at LREC-COLING)*. \*Equal contribution.
- 2024 Arnett, C.\*, Rivière, P. D.\*, **Chang, T. A.**, & Trott, S. Different tokenization schemes lead to comparable performance in Spanish number agreement. *Proceedings of the SIGMORPHON Workshop on Computational Research in Phonetics, Phonology, and Morphology (workshop at NAACL)*. \*Equal contribution.
- 2023 Arnett, C., **Chang, T. A.**, Michaelov, J., & Bergen, B. K. Crosslingual structural priming and the pre-training dynamics of bilingual language models. *3rd Multilingual Representation Learning Workshop (workshop at EMNLP)*. Extended abstract.
- 2020 **Chang, T. A.**, & Rafferty, A. N. Encodings of source syntax: Similarities in NMT representations across target languages. *Proceedings of the 5th Workshop on Representation Learning for NLP (workshop at ACL)*.

---

## OTHER WORK

---

- 2025      **Chang, T. A.** N-gram learning and pretraining dynamics in Transformer language models. *Doctoral dissertation, University of California San Diego, Department of Cognitive Science.*
- 2024      **Chang, T. A.,** Rajagopal, D., Bolukbasi, T., Dixon, L., & Tenney, I. Scaling training data attribution: The science of how training data influences LLM behavior. *People & AI Research, Google DeepMind, Medium Blog.*
- 2023      Thain, N., **Chang, T. A.,** Dixon, L., Croak, M., Meier-Hellstern, K., Tomanek, K., Hoffmann, J., & van Liemt, E. Detecting errors in chat bot outputs using language model neural networks. *United States patent application.* Work done at Google Research.
- 2020      **Chang, T. A.** Emergence of hierarchical syntax in neural machine translation. *Undergraduate thesis, Carleton College, Department of Cognitive Science.* With distinction.
- 2020      **Chang, T. A.** Topology of second order tensor fields. *Undergraduate thesis, Carleton College, Department of Mathematics and Statistics.*

---

## TEACHING

---

- 2021, 2025    **Teaching Assistant, UC San Diego**  
COGS188 AI Algorithms, Winter 2021  
COGS108 Data Science in Practice, Fall 2021, Winter 2025
- 2019      **Course Staff, Carleton College**  
CS254 Computability and Complexity, Winter 2019  
CS111 Introduction to Computer Science, Spring 2019

---

## AWARDS AND FELLOWSHIPS

---

- 2025      Google Spot Bonus, external multilinguality evals for Gemini 3 launch
- 2025      Spotlight Paper, NeurIPS 2025
- 2024      Outstanding Paper Award, EMNLP 2024
- 2020-2024    Graduate Prize Fellowship, UCSD Halicioğlu Data Science Institute
- 2020-2024    Glushko Travel and Research Award, UCSD Cognitive Science
- 2020      Roy O. Elveton Prize in Cognitive Science and Philosophy, Carleton College
- 2019      Google Spot Bonus, Google Geo Machine Perception
- 2019-2020    Patricia V. Damon Scholarship, Carleton College
- 2019      Phi Beta Kappa Third-Year Inductee, Carleton College
- 2018      Phi Beta Kappa First Year Prize, Carleton College
- 2017-2019    Dean's List, Carleton College
- 2017-2020    Stuebe Endowed Scholarship, Carleton College
- 2016-2020    National Merit Scholar, National Merit Scholarship Corporation

---

**SERVICE**

---

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reviewer   | <i>Proceedings of the National Academy of Sciences (PNAS)</i><br><i>Nature Communications</i> (Springer Nature)<br><i>Nature Scientific Reports</i> (Springer Nature)<br><i>Journal of Memory and Language</i> (JML)<br><i>Cognitive Systems Research</i><br><i>Artificial Intelligence Review</i><br><i>EPJ Data Science</i> (European Physical Journal)<br><i>BMC Medical Education</i> (BioMed Central)<br><br>ACL Rolling Review (ARR, Association for Computational Linguistics)<br>NeurIPS 2025-2026 (Conference on Neural Information Processing Systems)<br>ICLR 2025 (International Conference on Learning Representations)<br>CogSci 2022-2024, 2026 (Conference of the Cognitive Science Society)<br>ACL 2023-2024 (Meeting of the Association for Computational Linguistics)<br>FAccT 2023 (Conference on Fairness, Accountability, and Transparency)<br>EMNLP 2022-2024 (Conference on Empirical Methods in NLP) |
| 2026       | EMNLP (Conference on Empirical Methods in NLP),<br>Senior area chair, interpretability and analysis of models for NLP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2026       | Computational Developmental Linguistics (CDL) Workshop at ACL,<br>Co-organizer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2025, 2026 | Multilingual Representation Learning (MRL) Workshop at EMNLP,<br>Co-organizer, publication chair (2025), shared task co-lead                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2022-2023  | UCSD Graduate Application Mentorship Program, mentor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2022       | ICLR volunteer (International Conference on Learning Representations)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2020-2022  | ACL conference volunteer (Association for Computational Linguistics)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2020, 2022 | EMNLP conference volunteer (Empirical Methods in NLP)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |